EXCELLENCE

CLASS 11 - 2026-27

Physics | Lecture - 14

Basic Maths and Calculus

Applications of Differentiation and Maxima-Minima

By - Excellence Programme

18 September 2026

Topics to be covered

01 5-minute differentiation recap

02 Position, velocity and acceleration

03 Exact velocity–time graphs

04 Related rates: circle, square and sphere

05 Stationary points

06 Second derivative test

07 Maxima and minima practice

Rates of change

position $x(t) \rightarrow$ velocity $v = dx/dt$

velocity $v(t) \rightarrow$ acceleration $a = dv/dt = d^2x/dt^2$

stationary point: $dy/dx = 0$

related rates: differentiate both sides with respect to time

From position to a velocity graph

Differentiate the position function to obtain velocity.

Then plot v against t using its exact equation.

$x(t) \rightarrow v(t) = dx/dt \rightarrow v\text{–}t$ graph

Where $v = 0$, the object is instantaneously at rest.

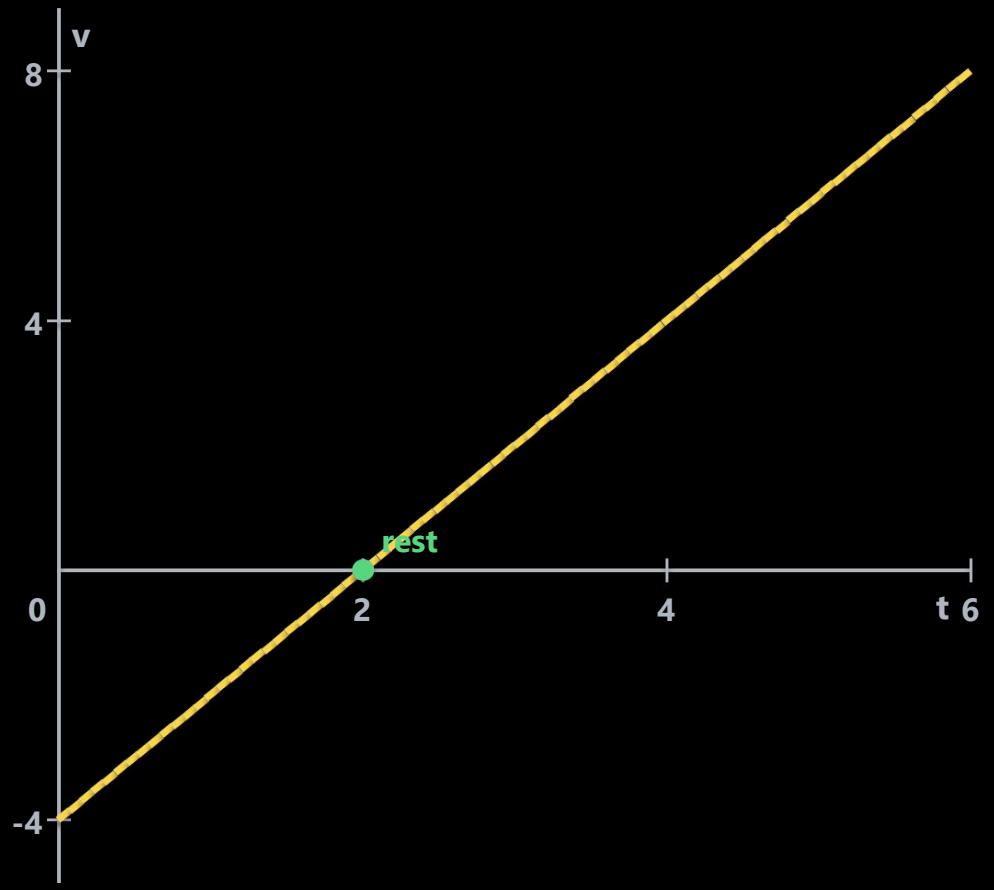
Motion: increasing velocity line

If $x = t^2-4t+8$, find velocity and draw the exact v–t graph.

Solution: Motion: increasing velocity line

Step 1: v = dx/dt = 2t-4

Step 2: v(0) = -4 and v = 0 at t = 2



Rough work / extra work

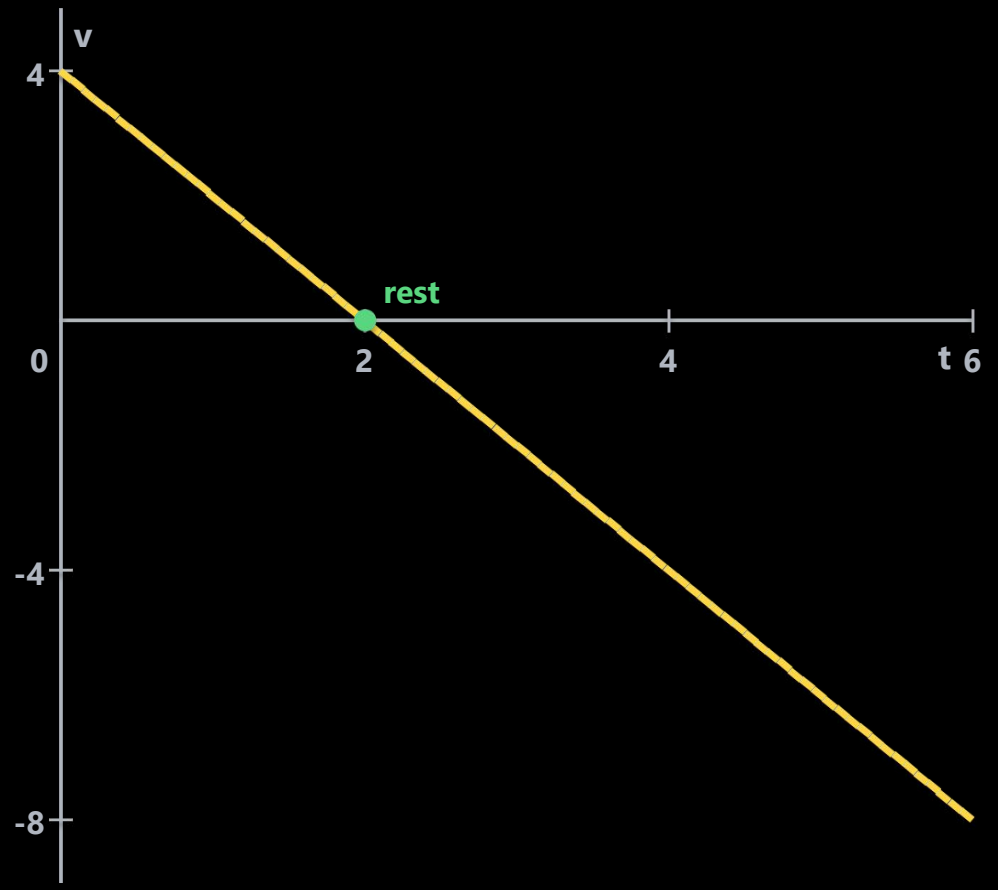
Motion: decreasing velocity line

If $x = -t^2+4t+8$, find velocity and draw the exact v–t graph.

Solution: Motion: decreasing velocity line

Step 1: v = dx/dt = -2t+4

Step 2: v(0) = 4 and v = 0 at t = 2



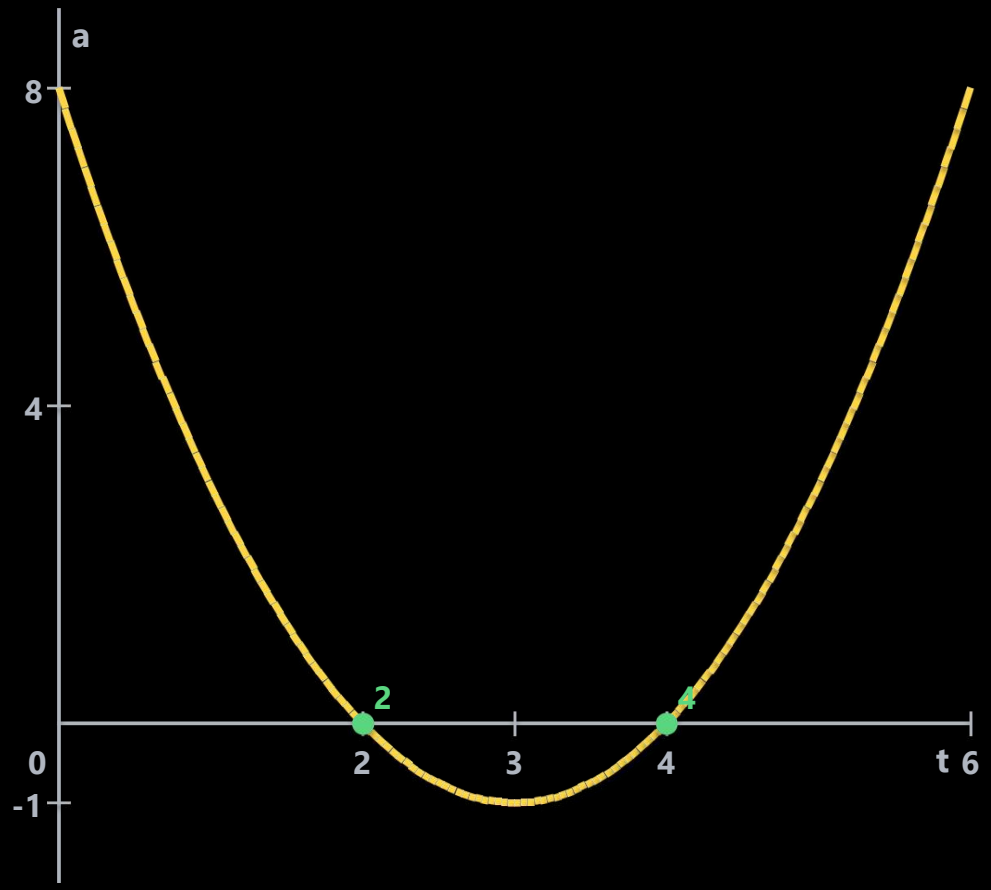
Acceleration from velocity

If $v = t^3/3 - 3t^2 + 8t$, find acceleration and draw the exact a–t graph.

Solution: Acceleration from velocity

Step 1: $a = dv/dt = t^2-6t+8$

Step 2: $a = (t-2)(t-4)$



Rough work / extra work

Changing x with time

If $y = x^2$ and x changes with time, express dy/dt in terms of x and dx/dt.

Solution: Changing x with time

Step 1: $dy/dx = 2x$

Step 2: $dy/dt = (dy/dx)(dx/dt)$

$dy/dt = 2x(dx/dt)$.

Changing angle with time

If $y = \sin\theta$ and θ changes with time, express dy/dt.

Solution: Changing angle with time

Step 1: $dy/d\theta = \cos\theta$

Step 2: $dy/dt = (dy/d\theta)(d\theta/dt)$

$dy/dt = \cos\theta(d\theta/dt)$.

Nested time dependence

If $y = \sin(t^2+2t)$, find dy/dt.

Solution: Nested time dependence

Step 1: Outer derivative: $\cos(t^2+2t)$

Step 2: Inner derivative: $2t+2$

$dy/dt = (2t+2)\cos(t^2+2t)$.

Rough work / extra work

Related rates

When dimensions change with time, geometrical quantities also change.

Write the geometry formula first, then differentiate with respect to time.

formula → differentiate with respect to t → substitute values with units

Substitute numerical values only after differentiating.

Circle, square and sphere formulae

circle: $A = \pi r^2 \rightarrow dA/dt = 2\pi r(dr/dt)$

square: $P = 4l \rightarrow dP/dt = 4(dl/dt)$

sphere: $V = (4/3)\pi r^3 \rightarrow dV/dt = 4\pi r^2(dr/dt)$

sphere: $S = 4\pi r^2 \rightarrow dS/dt = 8\pi r(dr/dt)$

Sphere volume rate

For $V = (4/3)\pi r^3$, express dV/dt in terms of r and dr/dt.

Solution: Sphere volume rate

Step 1: Differentiate with respect to t

$dV/dt = 4\pi r^2(dr/dt)$.

Rough work / extra work

Sphere surface-area rate

For $S = 4\pi r^2$, express dS/dt in terms of r and dr/dt.

Solution: Sphere surface-area rate

Step 1: $d(r^2)/dt = 2r(dr/dt)$

$dS/dt = 8\pi r(dr/dt)$.

Expanding circle

A circle's radius increases at $1/\pi$ m s^{-1}. Find dA/dt when $r = 4$ m.

Solution: Expanding circle

Step 1: $dA/dt = 2\pi r(dr/dt)$

Step 2: $= 2\pi(4)(1/\pi)$

$dA/dt = 8\ m^2\ s^{-1}$.

Growing square perimeter

The side of a square increases at 0.2 cm s^{-1}. Find the rate of increase of its perimeter.

Solution: Growing square perimeter

Step 1: $P = 4l$

Step 2: $dP/dt = 4(dl/dt) = 4(0.2)$

$dP/dt = 0.8\ \text{cm s}^{-1}$.

Rough work / extra work

Area change with radius

For a circle $A = \pi r^2$, find dA/dr.

Solution: Area change with radius

$dA/dr = 2\pi r$.

Inflating balloon

A spherical balloon's volume increases at 10 $cm^3\ s^{-1}$. Find dr/dt when r = 5 cm.

Solution: Inflating balloon

Step 1: $dV/dt = 4\pi r^2(dr/dt)$

Step 2: $10 = 4\pi(25)(dr/dt)$

$dr/dt = 1/(10\pi)$ cm s^{-1}.

Rough work / extra work

Growing square area

A square's side increases at 2 cm s^{-1}. Find dA/dt when the side is 10 cm.

Solution: Growing square area

Step 1: $A = l^2 \rightarrow dA/dt = 2l(dl/dt)$

Step 2: $= 2(10)(2)$

$dA/dt = 40\ cm^2\ s^{-1}$.

When is the particle at rest?

If $v(t) = t^2-3t$, when is the particle at rest?

Solution: When is the particle at rest?

Step 1: At rest: $v = 0$

Step 2: $t(t-3) = 0$

$t = 0$ s or 3 s.

Acceleration from displacement

If $s(t) = 3t^2-4t+2$, find acceleration at $t = 1$ s.

Solution: Acceleration from displacement

Step 1: $v = ds/dt = 6t - 4$

Step 2: $a = dv/dt = 6$

Acceleration = 6 m s^{-2}.

Rough work / extra work

Velocity from displacement

If $x(t) = 4t^3+3t^2-6t+1$, find velocity at $t = 2$ s.

Solution: Velocity from displacement

Step 1: $v = dx/dt = 12t^2 + 6t - 6$

Step 2: $v(2) = 48 + 12 - 6$

Velocity = 54 m s^{-1}.

Exponential position

If $x(t) = e^{2t}+5$, find velocity.

Solution: Exponential position

$v = dx/dt = 2e^{2t}$.

Rough work / extra work

Maxima and minima

At a smooth local maximum or minimum, the tangent is horizontal.

Step 1: solve dy/dx = 0 to find stationary points

A stationary point may be a maximum, minimum or neither. Test it using the second derivative.

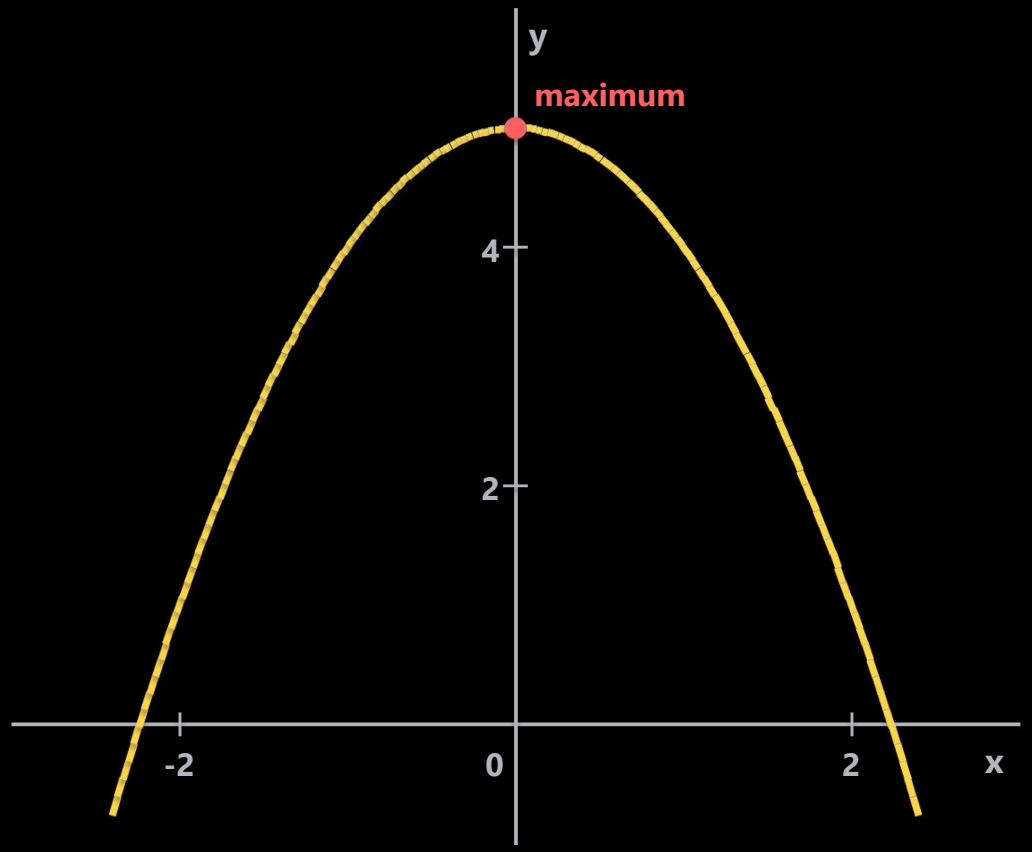
Second derivative test

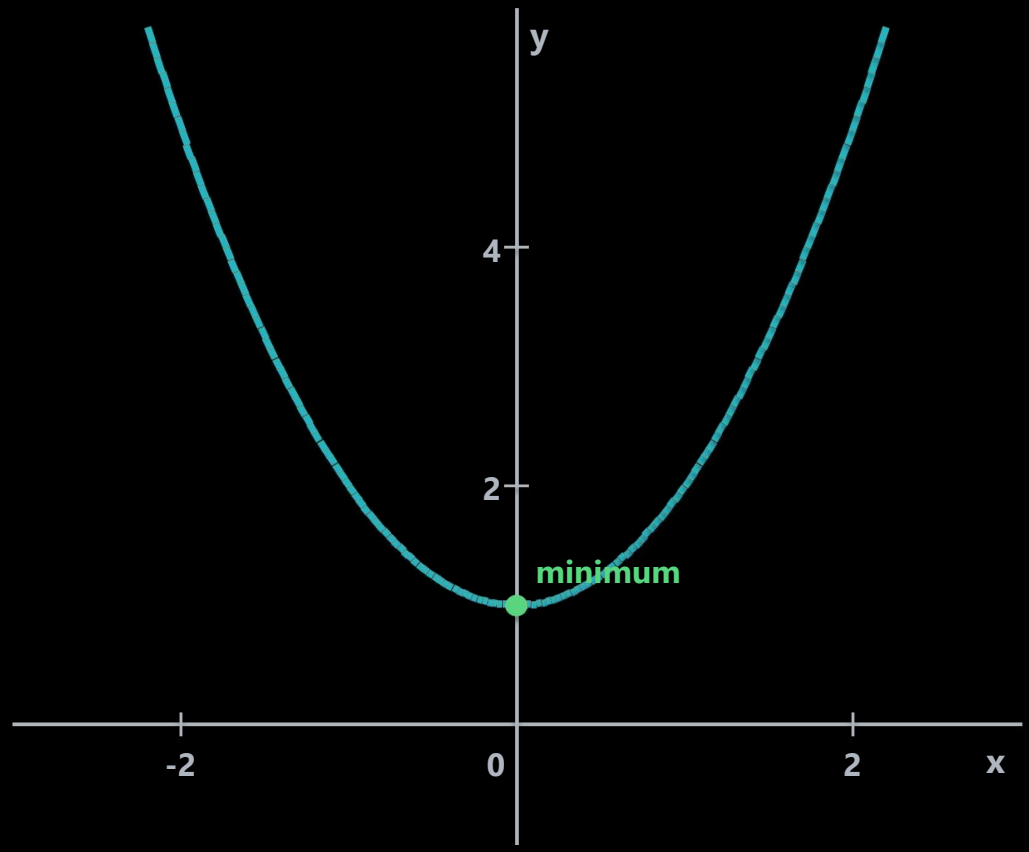
$$\frac{d^2y}{dx^2} > 0 \text{ at a stationary point} \rightarrow \text{local minimum}$$

$$\frac{d^2y}{dx^2} < 0 \text{ at a stationary point} \rightarrow \text{local maximum}$$

If $\frac{d^2y}{dx^2} = 0$, this test is inconclusive; another test is needed.

Maximum and minimum shapes





Minimum of a quadratic

Find the maxima and minima of $y = x^2+5$.

Solution: Minimum of a quadratic

Step 1: $dy/dx = 2x = 0 \rightarrow x = 0$

Step 2: $d^2y/dx^2 = 2 > 0 \rightarrow$ minimum

Minimum value $y = 5$ at $x = 0$; no maximum.

Rough work / extra work

Maximum of a quadratic

Find the maxima and minima of $y = 4x-x^2$.

Solution: Maximum of a quadratic

Step 1: $dy/dx = 4-2x = 0 \rightarrow x = 2$

Step 2: $d^2y/dx^2 = -2 < 0 \rightarrow$ maximum

Maximum value $y = 4$ at $x = 2$; no minimum.

Two stationary points

For $y = x^3-3x^2+4$, find its local maximum and minimum values.

Solution: Two stationary points

Step 1: $dy/dx = 3x(x-2) = 0 \rightarrow x = 0, 2$

Step 2: $d^2y/dx^2 = 6x-6$

Step 3: $x = 0$ gives maximum; $x = 2$ gives minimum

Local maximum = 4; local minimum = 0.

Maximum and minimum velocity

If $v = t^3-6t^2+12$, find its local maximum and minimum values.

Solution: Maximum and minimum velocity

Step 1: $dv/dt = 3t(t-4) = 0 \rightarrow t = 0, 4$

Step 2: $d^2v/dt^2 = 6t-12$

Local maximum $v = 12$ at $t = 0$; local minimum $v = -20$ at $t = 4$.

Rough work / extra work

Minimum by completing the test

Find the maxima and minima of $y = x^2-4x+5$.

Solution: Minimum by completing the test

Step 1: $dy/dx = 2x-4 = 0 \rightarrow x = 2$

Step 2: $d^2y/dx^2 = 2 > 0$

Minimum value $y = 1$ at $x = 2$; no maximum.

Cubic maxima and minima

For $y = x^3-12x^2+36x$, find the local maximum and minimum values.

Solution: Cubic maxima and minima

Step 1: $dy/dx = 3(x-2)(x-6) = 0 \rightarrow x = 2, 6$

Step 2: $d^2y/dx^2 = 6x-24$

Local maximum = 32 at x = 2; local minimum = 0 at x = 6.

Rough work / extra work

Applications of differentiation

$v = dx/dt$, $a = dv/dt = d^2x/dt^2$

related rates: differentiate the geometry formula with respect to t

stationary point: $dy/dx = 0$

$d^2y/dx^2 > 0$ minimum; $d^2y/dx^2 < 0$ maximum

Practice

1. Redraw all three motion graphs using their exact equations.

2. Memorise the related-rate formulae for a circle, square and sphere.

3. Redo all maxima–minima questions without seeing the answers.

4. Include units in every rate answer.

Never call dy/dx = 0 alone proof of a maximum or minimum.